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Reducing NO_x Emissions in Ammonia Combustors

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Ammonia continues to attract growing interest as a carbon-neutral replacement fuel, motivating numerous research efforts toward understanding fundamental ammonia combustion characteristics. A major challenge for the use of ammonia is the development of combustor technologies for mitigating potentially high NO_x emissions from the fuel-bound nitrogen chemical pathways to acceptable levels. Our work focuses on a staged rich-burn/quick-quench/lean-burn (RQL) combustor architecture for minimizing the NO_x emission levels through burning fuel-rich in the primary stage to form combustion products containing significant levels of hydrogen in addition to nitrogen and water with minimal NO_x formation. The subsequent quench and burnout stages of the combustor must then quickly burn residual hydrogen with flame-temperatures moderated by nitrogen and water forming in the first stage. Chemical reactor network (CRN) modeling was used to understand and identify optimal stoichiometry and residence times in each stage for minimizing NO_x emissions and to quantify pressure and temperature effects. Reducing the overall NO_x emissions requires relatively long residence times in the primary stage to achieve near equilibrium NO levels due to kinetically controlling processes. For conditions relevant to gas turbines (e.g., 30 atm), our work indicates that NO_x emissions below 20 ppm are theoretically achievable in a staged RQL combustor architecture. However, these emission predictions significantly depend on the accuracies of currently available chemical kinetic mechanisms which have not been extensively validated under elevated pressure and temperature conditions relevant to gas turbines. [DOI: 10.1115/1.4069539]

Keywords: ammonia combustion, NO_x emission, chemical reactor network modeling

1 Introduction

Ammonia-fueled combustors are increasingly explored as a means to reduce carbon emissions in gas turbines, given ammonia's potential as a carbon-free alternative to hydrocarbon fuels. However, one of the key challenges associated with the combustion of ammonia in gas turbines is the emission of nitrogen oxides (NO_x), which in the atmosphere react with water vapor and solar radiation to form nitric acid, a component of acid rain, and ground level ozone, a component of smog. To achieve key performance metrics such as efficiency, ultralow NO_x emissions, and stability in both low- and high-power conditions, the *rich-burn/quick-quench/lean-burn* (RQL) combustor design for aircraft turbines is firmly established [1,2]. Although the RQL design was initially developed for hydrocarbon fuels and their blends with low heating value fuels [3–7], it is now increasingly being explored as a potential option for ammonia combustion [8–10].

The two most significant parameters that influence both power performance requirements and NO_x formation in an NH_3-fired gas turbine are the equivalence ratio and the residence time of the fuel–air mixture. In the RQL design, combustion is initiated in the fuel-rich primary zone (front-end) and the fuel–air mixture is quickly reacted and diluted (quenched) with additional airflow in the secondary zone (downstream section) of the combustor to reach the fuel-lean conditions at the combustor exit. The concept underlying the RQL combustor for NH_3 combustion is depicted in Figs. 1(*a*) and 1(*b*), which are described further below in the context of emissions and how NH_3 chemical kinetics determine the approach to equilibrium levels. All simulation results shown in these figures for NH_3 combustion are performed using the Stagni mechanism [11].

The initial fuel-rich combustion process, denoted as "rich burn" in Fig. 1(*a*), effectively suppresses NO_x formation by limiting the availability of oxygen for NO_x formation, resulting in oxygen preferentially being consumed to form the low-energy product species H_2O rather than NO_x. To illustrate the important effects of equivalence ratio and residence time, Fig. 1(*a*) shows the computed peak NO levels in adiabatic freely propagating flames for conditions relevant to gas turbine combustors—35 atm and initial reactant temperature of 850 K. The peak NO concentrations within the flame zones at these conditions drop precipitously with increasing equivalence ratio (ϕ) on the rich side, and increasing residence time (τ) in the postflame region toward lower equilibrium levels. For a flame residence time of about 40 ms from the maximum heat release location, the local NO concentration within the postflame zone already drops to near equilibrium levels.

Figure 1(*b*) plots the variation for the NH_3 concentration with equivalence ratio in comparison to equilibrium NO and NH_3 levels. As the equivalence ratio increases, the NO levels decrease, whereas the NH_3 levels in the postflame zone increase with rising ϕ. At a residence time of 40 ms in the postflame region, the local NH_3 levels

Turbo Expo: Turbomachinery Technical Conference & Exposition (GT2025), June 16–20, 2025. GT2025.
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Manuscript received July 3, 2025; final manuscript received July 29, 2025; published online October 8, 2025. Editor: Jerzy T. Sawicki.

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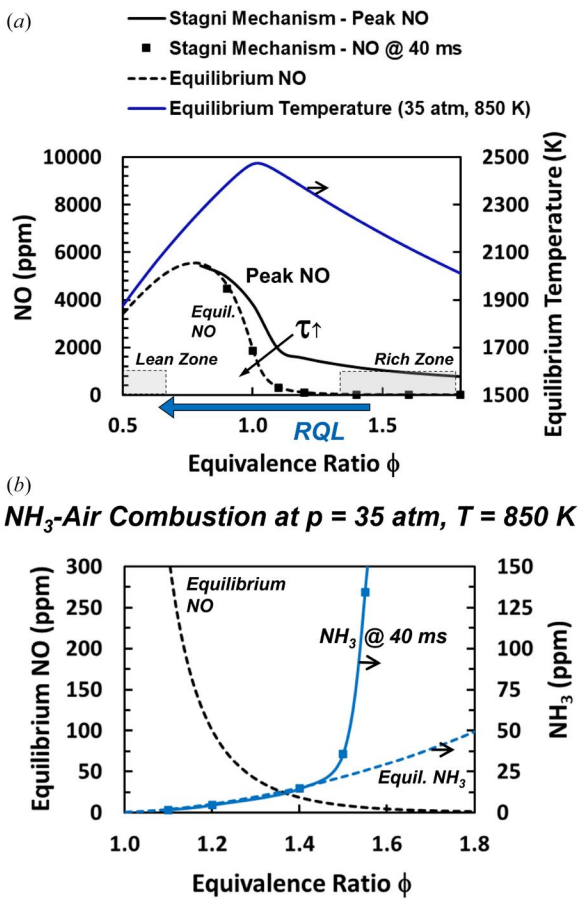

Fig. 1 Illustration of pollutant species NH_3 and NO at varying equivalence ratios, showing basis for RQL combustion of pure NH_3 fuel with low NO_x emissions. (*a*) Concentration of peak NO in flame at <1 ms from maximum heat release location (solid line); NO concentration in flame at 40 ms from maximum heat release location (solid square symbols); and NO concentration at equilibrium (dashed line). Equilibrium temperature (adiabatic flame temperature) is shown using the right-hand axis. (*b*) Concentration of NH_3 illustrated using the right-hand axis, with the solid line indicating NH_3 concentration after 40 ms from maximum heat release location, and the dashed line indicating NH_3 concentration at equilibrium. All values are calculated by the authors using the Stagni mechanism [11] for ammonia combustion, for ammonia–air freely propagating premixed flames at 850 K inlet temperature and 35 atm pressure.

correspond closely with equilibrium levels until about $\phi = 1.5$, beyond which local NH_3 levels significantly exceed equilibrium concentrations. Subsequently, the optimum combustor performance in terms of overall equivalence ratio for lowest emissions manifests from the trade between NO reduction and NH_3 slippage, and Fig. 1(*b*) shows that this optimum occurs in the vicinity of $\phi = 1.4$. Consequently, in the fuel-lean secondary zone for RQL-type architectures, NO_x concentrations rise because of elevated temperatures and the presence of unburned NH_3 along with abundance of O/H radicals and available oxygen. Additionally, the emissions in both the primary and secondary zones are also affected by the residence time of the fuel–air mixture. While the optimal operating conditions for low NO_x emissions in RQL-type, gas turbine combustors using hydrocarbon fuels are relatively well established, the ideal conditions for NH_3-fired systems to minimize NO_x emissions—such as equivalence ratio, pressure, exit temperature, and residence time—are an area of active research. A deeper understanding of ammonia combustion kinetics is essential to determine the optimal conditions for reducing NO_x emissions in high-pressure turbine operations [10,12,13].

The chemical kinetic studies and validation of RQL-staged combustion strategy in terms of pollutant predictions can be accomplished using the widely adopted chemical reactor network (CRN) method. Several recent studies [14–16] have confirmed the effectiveness of the RQL design approach for the combustion of NH_3 and its blends with other fuels through the application of CRN modeling. Gubbi et al. [14] used a two-stage reactor network model comprised of freely propagating flame models in combination with perfect mixers to evaluate the theoretical minimum achievable NO emissions for constrained combustor exit H_2 values. By determining the minimum NO concentrations, they identified the optimal fuel-rich stage equivalence ratio (ϕ_r) and fuel-lean stage residence time for a given total residence time. Their studies predicted a monotonic increase in NO concentrations at the combustor exit with increasing fuel-lean stage residence times, at all temperatures. Additionally, their research indicates that an increase in pressure will monotonically decrease total NO levels at the combustor exit for a given temperature.

Li et al. [15,16] used two-stage CRN modeling to elucidate the effects of oxygen enrichment and hydrogen addition on NO_x emissions for NH_3 combustion in a typical E-class turbine at a given design pressure, as well as inlet and outlet temperatures. Their studies showed that the rich-stage equivalence ratio is the primary factor in minimizing NO_x emissions. The concentrations of NH_i and OH/H radicals in the rich-stage determine the dominant mechanisms for NO formation and destruction. However, the aforementioned studies lack a systematic and parametric analysis of the effects of residence times, pressure, temperature, and NH_3-slip on NO_x formation under gas turbine conditions. Moreover, they lack essential reaction path and sensitivity analyses for a range of gas turbine design parameters required to optimize the design for ammonia combustion. These chemical kinetic analyses are crucial for devising strategies to enhance turndown and reduce NO_x emissions in ammonia–air combustors.

This study aims to address this gap by further investigating NO_x emission control mechanisms in RQL combustors using CRN modeling. It involves the CRN simulations for varied combustion parameters relevant to gas turbines to investigate effects of varying equivalence ratio and residence times in the rich and lean stage, exit temperature, and design pressure on the NO_x concentration. Additionally, rate-of-production (ROP) and sensitivity analyses were performed to identify the major reaction pathways critical to production or consumption of NO_x and NH_3 at critical instants in the reactor network.

2 Modeling

The chemical reactor network modeling method is widely used for the chemical kinetic studies of NO_x emissions in RQL staged NH_3-fired gas combustor architectures. CRN modeling offers an optimum approach with the detailed chemical kinetics of complex combustion processes and consequently, good accuracy in predicting NO_x emissions in gas turbine combustors. In this paper, a CRN modeling study on the NO_x emission characteristics for a two-stage RQL ammonia–air combustion was conducted using CANTERA [17] and PYTHON interface. CANTERA is an open-source flame/chemical kinetics software tool. Simulations for multiple design cases were performed for different combustion parameters to estimate the overall NO_x conversion rate under varying conditions.

2.1 Chemical Reactor Network Model. The CRN model consists of a fuel-rich stage and a fuel-lean stage combustion regions, as illustrated in Fig. 2. The model consists of a pair of perfectly stirred (PSR) and plug-flow (PFR) reactors connected in series for both the fuel-rich and fuel-lean stages. For the fuel-rich stage, a PSR was used to represent the fuel–air mixing and flame anchoring region, whereas a PFR was used to represent the combustion region downstream of the flame anchoring zone. In the case of the fuel-lean stage, quick mixing of the secondary air stream with fuel-rich stage combustion products was modeled using

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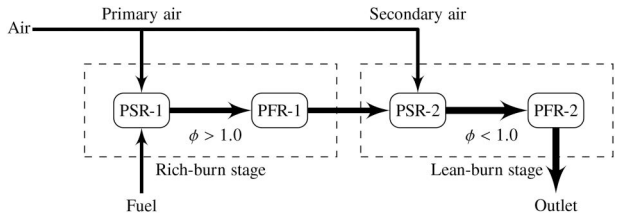

Fig. 2 The CRN model for a two-stage RQL combustor

a PSR, and the postmixing combustion was simulated through PFR modeling. Using CANTERA's `IdealGasReactor` construct, an adiabatic reactor model at constant pressure was used to simulate the PSR, while the PFR was modeled using `ConstantPressureReactor` class object which is an adiabatic constant-pressure system. To incorporate the thermodynamic properties and reaction rates in the calculation of evolution of species, a mechanism file in. `YAML` format is required for the CANTERA transient simulations. Separate gas phase "objects" (using the same mechanism) were created for the fuel-rich and fuel-lean stages to calculate the concentration, rate of production, rate of reaction, and other thermodynamic properties of species at each instant of time. Along with the design residence times and inlet temperature and pressure, a range of fuel-rich stage equivalence ratios were selected as input parameters, and the amount of secondary air (and fuel-lean stage equivalence ratio, ϕ_l) was determined by iteration to maintain the given design combustor outlet or exhaust temperature.

2.2 Chemical Kinetic Mechanism. A comparative CRN analysis was conducted through simulation of a particular set of input parameters using various published N/H mechanisms. The four published ammonia mechanisms examined here include Glarborg [18], Powell [19], Mei [20], and Stagni [11]. The results of this analysis are depicted in Fig. 3 by plotting NO_x as a function of global residence time (τ_g). The NO_x concentration at the exit of PFR-2 in these simulations ranges from 12.12 ppm to 13.05 ppm for $\phi_r = 1.45$. In this paper, NO_x is defined as the total concentration of NO, NO_2, and N_2O. All the NO_x levels have been corrected to dry 15 % vol. O_2. The time scale for NO_x reduction at early time in PFR-1 is kinetically driven; as a result, significant variations are expected in this region across different N/H chemical kinetic mechanisms in the literature, as well as in their respective relaxation rates toward equilibrium. Previous work has [21] shown that the Stagni mechanism gives reasonable predictions for laminar flame speeds and counterflow extinction strain rates for NH_3–air flames at elevated pressure. Therefore, the detailed Stagni mechanism with 31 species and 203 reactions was selected here to study the NO_x emission characteristics and reaction pathways analyses of NH_3–air combustion.

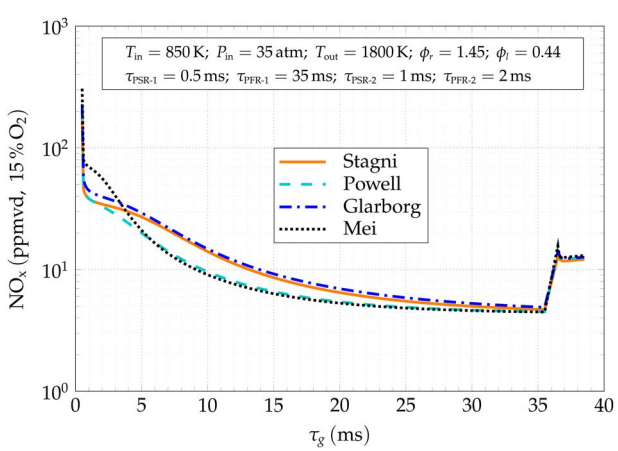

Fig. 3 Prediction of NO_x from different mechanisms for baseline conditions

Table 1 Combustion parameters used in the CRN modeling

Study set	τ_{total} (ms)	τ_{PSR-1} (ms)	τ_{PFR-1} (ms)	τ_{PSR-2} (ms)	τ_{PFR-2} (ms)	P_{in} (atm)	T_{in} (K)	T_{out} (K)	ϕ_r
A	38.5	0.5	35	1	2	35	850	1800	1.0–1.8
B	11.5–41.5	0.5	5–35	1	2–5	35	850	1700	1.0–1.8
C	11.5–41.5	0.5	5–35	1	2–5	35	850	1800	1.0–1.8
D	11.5–41.5	0.5	5–35	1	2–5	35	850	1900	1.0–1.8
E	11.5–41.5	0.5	5–35	1	2–5	20	850	1800	1.0–1.8
F	11.5–41.5	0.5	5–35	1	2–5	30	850	1800	1.0–1.8
G	11.5–41.5	0.5	5–35	1	2–5	40	850	1800	1.0–1.8

Highlighted areas are the key parametric changes.

2.3 Model Parameters. A parametric study was performed over a range of conditions relevant to gas turbine cycles, including operating pressures, temperatures, and residence times characteristic of a typical aeroderivative cycle. Table 1 provides the combustion parameters used for simulating the various study sets considered in this study. The *baseline* combustor inlet conditions were selected as $P_{in} = 35$ atm and $T_{in} = 850$ K to be representative of an aeroderivative gas turbine [22], and the baseline combustor exit temperature was selected as $T_{out} = 1800$ K (with ± 100 K case variations) to be representative of low-NO_x combustor design practice in both aeroderivative and frame engines [23]. In addition, the mixing residence times for flame-anchoring ($\tau_{PSR-1} = 0.5$ ms) and for secondary-air mixing ($\tau_{PSR-2} = 1.0$ ms) were selected to be consistent with aeroderivative engine design practice [2]. By definition, each perfectly stirred reactor model assumes an ideal steady-state reactor with instantaneous and perfectly uniform concentrations throughout the control volume. In contrast, a plug-flow reactor represents a one-dimensional, steady-state ideal reactor with varying concentrations along the spatial coordinate in our CRN model. In all results reported here, a residence time of 0 ms corresponds to the beginning of PSR-1.

Initial simulations suggest that the optimal NO_x emissions at the RQL system outlet are significantly influenced by the residence time in the PFR-1, as shown in Fig. 4. The trend lines in the figure correspond to a total of 16 simulations for a range of total residence times. These simulation lines are grouped based on $\tau_{PFR-1} =$ 5 ms, 10 ms, 15 ms, and 20 ms under baseline conditions, with $\tau_{PFR-2} = 5$ ms, resulting in overlapping plotted lines within each group. These initial and subsequent simulations also indicate that the

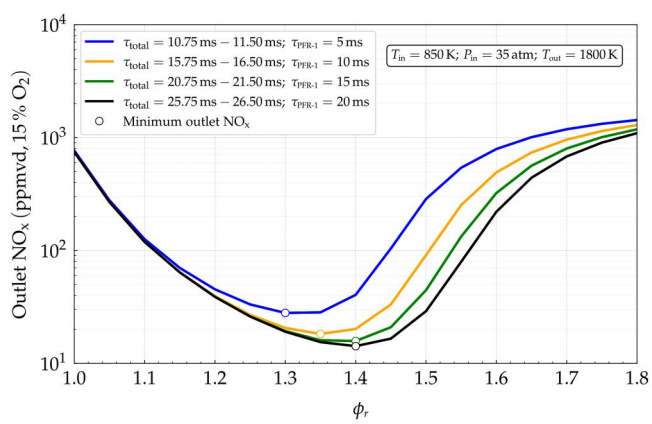

Fig. 4 Effects of residence times and rich-stage equivalence ratio on prediction of NO_x at the outlet. The simulations are presented for a range of total residence times under baseline conditions, grouped by τ_{PFR-1}=5 ms, 10 ms, 15 ms, and 20 ms.

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Table 2 Summary of results from selected simulations

Case	τ_{total} (ms)	τ_{PFR-1} (ms)	τ_{PFR-2} (ms)	P_{in} (atm)	T_{out} (K)	ϕ_r	ϕ_l [c]	Rich stage[a] NH_3 [d] (ppm)	Rich stage[a] NO_x [e] (ppm)	Lean stage[b] NH_3 [d] (ppb)	Lean stage[b] NO_x [e] (ppm)
A1	38.5	35	2	35	1800	1.45	0.45	17.5	4.7	0.023	12.1
A2	38.5	35	2	35	1800	1.80	0.45	6384.3	2.6	0.013	895.4
A3	38.5	35	2	35	1800	1.10	0.45	1.7	120.4	0.021	118.3
C1	11.5	5	5	35	1800	1.30	0.45	12.6	23.2	0.022	28.3
C2	41.5	35	5	35	1800	1.45	0.45	17.5	4.6	0.023	12.9
D1	41.5	35	5	35	1900	1.45	0.51	17.5	4.6	0.073	19.4
D2	38.5	35	2	35	1900	1.45	0.51	17.5	4.6	0.073	15.3
E1	11.5	5	5	20	1800	1.35	0.45	14.7	36.2	0.017	40.0
E2	41.5	35	5	20	1800	1.45	0.45	10.5	6.5	0.018	11.9
G1	11.5	5	5	40	1800	1.30	0.45	13.7	20.3	0.024	26.2
G2	41.5	35	5	40	1800	1.40	0.45	15.4	6.0	0.024	13.4

Highlighted areas are the key parametric changes.
[a]At the end of fuel-rich stage.
[b]At the end of fuel-lean stage.
[c]Lean-stage equivalence ratio is same as global equivalence ratio.
[d]Wet basis.
[e]15% O_2 dry basis.

optimum (corresponding to minimum NO_x emission) rich-stage equivalence ratio increases with total residence time, ranging from 1.30 to 1.45. Therefore, a range of residence time for PFR-1 was selected to investigate its effect on NO_x concentration in additional simulations. The range of conditions for these additional simulations are shown in Table 1.

3 Results and Discussion

This section systematically presents the simulation results, highlighting the influence of various operational parameters on NO_x emissions in the RQL combustor. Parameters such as equivalence ratio, design pressure, exit or outlet temperature, and residence time are examined to determine the contributions to NO_x formation.

Table 2 summarizes the results of selected simulations. In all cases, the residence times for PSR-1 and PSR-2 are 0.5 ms and 1.0 ms, respectively. Additionally, in all cases, the concentration of NH_3 decreases significantly from the rich-stage to the lean-stage. Cases A1 and A2 illustrate the considerable increase in NH_3-slip and NO_x concentrations in the rich and lean stages, respectively, due to an increase in the rich-stage equivalence ratio from 1.45 to 1.80. In cases C1 to G2, the listed ϕ_r values are the optimal equivalence ratio ($\phi_{r,opt}$) values yielding the corresponding minimum NO_x emission values. The effects of exit temperature and residence times on NH_3-slip and NO_x concentration are demonstrated by cases C1 to D2. Cases C1 and C2 show that, at the same exit temperature, increasing the residence time of PFR-1 reduces the NO_x concentration at the exit of PFR-2. A comparison of cases C2 and D1 reveals that both NO_x and NH_3-slip increase at the exit of PFR-2 with a rise in exit temperature. Note that the NH_3 concentrations at the lean-stage exit are negligible for all cases and are reported in the table as parts per billion. Furthermore, increasing the residence time of PFR-2 leads to a higher NO_x concentration at the combustor exit, as shown by the data for cases D1 and D2. A comparison of cases E1 and E2 shows that, at the same design pressure, NO_x concentration at the combustor exit decreases with an increase in the residence time of PFR-1. The impact of pressure and residence times on NH_3-slip and NO_x concentration is illustrated by cases E1 to G2. For a given total residence time, increasing the design pressure leads to a significant decrease in exit NO_x concentration for short τ_{PFR-1}, as shown by the comparison of cases E1 and G1. By increasing the τ_{PFR-1} from 5 ms to 35 ms, which in turn increase the total residence time from 11.5 ms to 41.5 ms, the NO_x emission levels of E2 and G2 are substantially reduced in comparison to E1 and G1. However, for E2 and G2, the increase in operating pressure results in a minor increase in outlet NO_x concentration. A more detailed discussion on these trends is provided in Sec. 3.3.

In addition to the above parametric variations, this study will also examine the rate of production and consumption analysis, sensitivity analysis, and the role of reaction pathways that influence NO_x formation and NH_3 concentration under various operating conditions.

3.1 Dependence of Rich-Stage Equivalence Ratio. To investigate the impact of the rich-stage equivalence ratio on NO_x emissions, ϕ_r was varied between 1.0 and 1.8, while the combustor exit temperature was maintained at a constant value by adjusting the secondary air flow into the second stage. Figure 5 shows the variation of NO_x and NH_3 concentrations throughout the chemical reactor network as a function of the global residence time for the baseline case for selected rich-stage equivalence ratios. Again, all the NO_x emission levels have been corrected to dry 15 % vol. O_2. At the start of the PFR-1 reactor, a rapid decrease in NO_x concentration is observed for all rich-stage equivalence ratios, with the initial decay rate increasing as the equivalence ratio is increased. In contrast, while a rapid decrease in NH_3 concentration is also observed in PFR-1 for all rich-stage equivalence ratios, the initial NH_3 decay rate decreases as the equivalence ratio is increased. Figures 5(*a*) and 5(*b*) show a rapid increase in NO_x concentrations and a decrease in NH_3 concentrations at the end of PSR-2, respectively, in the reactor network across all ϕ_r, following the

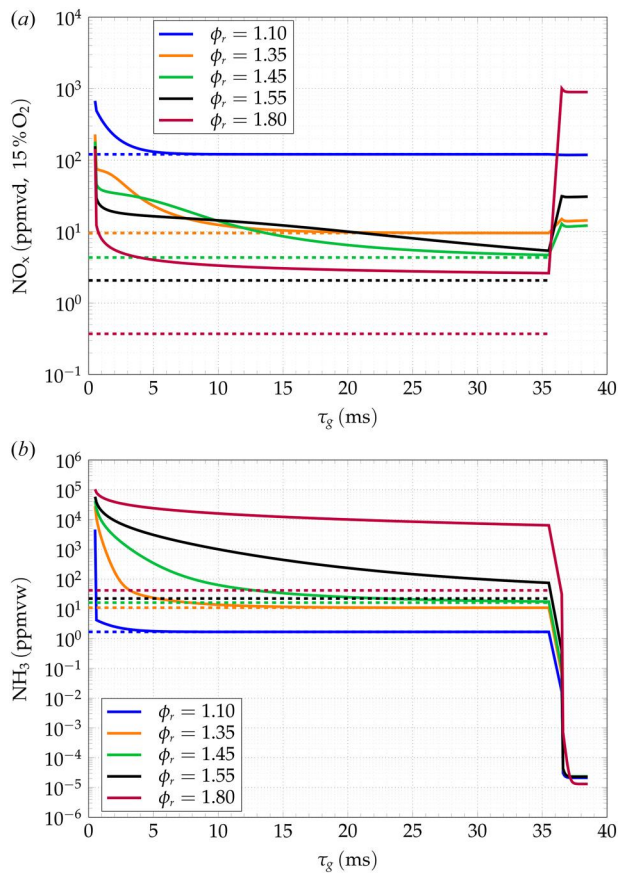

Fig. 5 Evolution of NO_x and NH_3 with global residence time in the combustor for baseline conditions (P_{in}=35 atm, T_{in}=850 K, T_{out}=1800 K, τ_{PSR-1}=0.5 ms, τ_{PFR-1}=35.0 ms, τ_{PSR-2}=1.0 ms, and τ_{PFR-2}=2.0 ms). The dashed lines show the equilibrium NO_x or NH_3 for the corresponding rich-stage equivalence ratio. (*a*) Evolution of NO_x with global residence time and (*b*) evolution of NH_3 with global residence time.

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introduction of secondary air. For the conditions examined, the maximum and minimum concentrations of NO_x at the combustor exit were observed at $\phi_r = 1.80$ and $\phi_r = \phi_{r,opt} = 1.45$, respectively. The increased NO_x formation for $\phi_r > \phi_{r,opt}$ can be attributed to increased NH_3-slip and fuel NO_x pathways in the fuel-lean stage. The dashed lines, which extend only to the end of the fuel-rich stage ($\tau_g = 35.5$ ms) for each case, represent the equilibrium NO_x or NH_3 for the rich-stage equivalence ratio. These equilibrium NO_x emission levels have also been corrected to dry 15 % vol. O_2. As demonstrated in Fig. 5(*b*), for $\phi_r > 1.5$, the approach to equilibrium for NH_3 concentration is considerably slower than for $\phi_r < 1.5$. Figure 5(*a*) shows that in the NO_x-optimal case ($\phi_r = 1.45$), NO_x levels decline rapidly during the fuel-rich stage as the τ_g in PFR-1 increases, eventually approaching equilibrium NO_x concentrations by the end of PFR-1. Additionally, the NO_x concentration remains nearly constant between residence times of approximately 25 ms and 35.5 ms in PFR-1. This suggests that the residence time of PFR-1 can be reduced for the optimum case for minimizing NO_x emissions. For $\phi_r < 1.45$, the NO_x concentrations for each ϕ_r tend to quickly approach their equilibrium values as τ_g increases.

3.2 Dependence of Exit Temperature. Simulations were carried out under baseline conditions to examine the effect of changing the combustor exit temperature on NO_x concentration. Figure 6 illustrates the NO_x emission levels for the combination of τ_{PFR-1} and τ_{PFR-2} at the outlet temperatures of 1700 K, 1800 K, and 1900 K. For each combination of τ_{PFR-1} and τ_{PFR-2}, the residence times of PSR-1 and PSR-2 were fixed at 0.5 ms and 1.0 ms, respectively. The figure presents a compilation of several individual graphs. In each subplot, corresponding to a given τ_{PFR-1}, the optimal NO_x is displayed as a function of τ_{PFR-2}, which varies from 0.05msto 5 ms. The optimum equivalence ratio for Fig. 6 ranges from 1.30 to 1.45 across all the conditions and residence times shown. The lean-stage or overall equivalence ratios corresponding to $T_{out} = 1700$ K, 1800 K, and 1900 K are 0.38, 0.45, and 0.51, respectively. The relatively small increase in ϕ_l is attributed to the decrease in the amount of secondary air added to the lean stage, which increases the exit temperature. The figure illustrates that for all three exit temperatures, NO_x concentrations exhibit a slight initial decline, followed by an increase, as the residence time of PFR-2 increases. The rise in NO_x concentration over time is most significant for $T_{out} = 1900$ K, whereas for the other two temperatures (1700 K and 1800 K), the rise is minimal. These results suggest that the residence time in PFR-2 has little to no impact on NO_x concentration for $T_{out} < 1800$ K. NO_x formation at $T_{out} > 1800$ K is driven by the Zel'dovich mechanism which is controlled in large part by the high activation energy rate-limiting reaction $N_2 + O \rightleftharpoons NO + N$. An increase in temperatures enhances the O/H radical pool by accelerating the temperature-sensitive chain branching reactions. As a result, elevated NO_x emission levels are observed in PFR-2 as the residence time increases. Furthermore, the HNO intermediate pathway serves as the primary route for NO formation in NH_3–air flames under all conditions [12]. HNO is primarily converted to NO through its reactions with H, OH, and O_2 as well as via thermal dissociation.

3.3 Dependence of Design Pressure. To evaluate the effect of varying the combustor design pressure, simulations were conducted under baseline conditions with adjustments to the pressure to examine its impact on NO_x concentration. Figure 7 presents the NO_x emission levels for the combination of τ_{PFR-1} and τ_{PFR-2} at design pressures of 20 atm, 30 atm, and 40 atm. For each combination of τ_{PFR-1} and τ_{PFR-2}, the residence times of PSR-1 and PSR-2 were fixed at 0.5 ms and 1.0 ms, respectively. The figure presents a compilation of several individual graphs. In each subplot, corresponding to a given τ_{PFR-1}, the optimal NO_x is displayed as a function of τ_{PFR-2}, which varies from 0.05 ms to 5 ms. The optimum equivalence ratio for Fig. 7 ranges from 1.30 to 1.45 across all the conditions and residence times shown. As the residence time in PFR-2 increases, the NO_x concentration initially shows a slight decrease, followed by a gradual rise with increase in τ_{PFR-2}. The figure indicates that at shorter τ_{PFR-1}, NO_x emissions decrease with increasing design pressure. However, for longer τ_{PFR-1}, a crossover occurs around 25–30 ms, after which NO_x emissions become higher at greater pressures. For $\tau_{PFR-1} > 25$ ms, the NO_x concentrations do not vary significantly across the simulated design pressure range. The graph illustrates the importance of elevated pressure for reducing NO_x emissions in the fuel-rich stage for shorter PFR-1 residence times below about 20 ms. The O/H radical pool is reduced with increasing pressure due to the three-body pressure sensitive reactions $H + OH + M \rightleftharpoons H_2O + M$ and $H + O_2\ (+M) \rightleftharpoons HO_2\ (+M)$. The former reaction is a chain terminating reaction, whereas the latter results in the formation of less active HO_2 radicals. Consequently, the active O/H radical pool is reduced, leading to a net decrease in NO production at high pressures, while NO

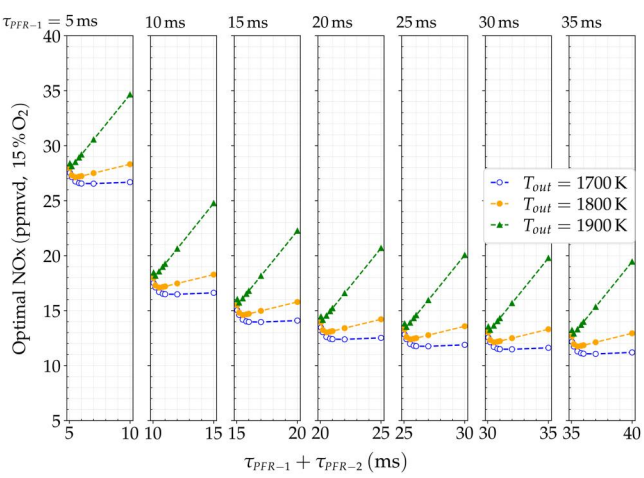

Fig. 6 Effects of exit temperature on optimal NO_x levels for different combinations of τ_{PFR-1} and τ_{PFR-2}. The figure shows a collection of several individual graphs, where in each subplot, τ_{PFR-2} varies from 0.05msto 5 ms for a given value of τ_{PFR-1}, indicated at the left edge of each subplot. τ_{PFR-1} ranges from 5 ms to 35 ms across the subplots. The conditions are P_{in}=35 atm, T_{in}=850 K, τ_{PSR-1}=0.5 ms, and τ_{PSR-2}=1.0 ms.

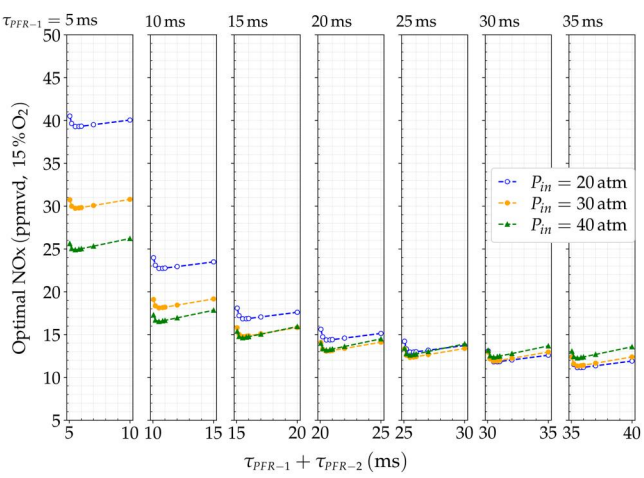

Fig. 7 Effects of pressure variation on optimal NO_x levels for different combinations of τ_{PFR-1} and τ_{PFR-2}. The figure shows a collection of several individual graphs, where in each subplot, τ_{PFR-2} varies from 0.05 ms to 5 ms for a given value of τ_{PFR-1}, indicated at the left edge of each subplot. τ_{PFR-1} ranges from 5 ms to 35 ms across the subplots. The conditions are T_{in}=850 K, T_{out}=1800 K, τ_{PSR-1}=0.5 ms, and τ_{PSR-2}=1.0 ms.

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consumption rates by NH_i species remain high [12]. As the residence time in PFR-1 increases, the net production rate of NO_x increases under fuel-lean conditions and elevated temperatures. This increase is attributed to the combined effects of increased concentrations of O/H/OH radicals, the activation of Zel'dovich mechanism pathways, and the accelerated reaction rates at higher pressures. These effects account for the higher concentration of NO_x at 40 atm compared to 20 atm at longer residence times.

3.4 Rate-of-Production Analysis. A local reaction flow analysis examines the formation and destruction of species at specific points, either in time for time-dependent problems or in space for steady, spatially dependent problems. To examine the primary NO_x reaction pathways contributing to its production or destruction in both the rich and lean burn stages, net ROP analyses were conducted to identify the dominant NO_x reaction steps. As NO is the primary contributor (with N_2O being the second most significant contributor) to NO_x concentrations in all simulations, the ROP analysis focuses primarily on NO for brevity. Examples of proportions of NO_x constituents are given at the end of Sec. 3.5. A rate-of-production analysis considers the percentage contributions of different reactions to the formation or destruction of a particular chemical species [24]. The normalized production contribution of a given reaction i to a particular species j for a mechanism that considers N species and L elementary reactions is given by

$$C_{ji}^{\mathrm{p}} = \frac{\max(\nu_{ji} q_i, 0)}{\sum_{i=1}^{L} \max(\nu_{ji} q_i, 0)} \tag{1}$$

The normalized destruction contribution is given by

$$C_{ji}^{\mathrm{d}} = -\frac{\min(\nu_{ji} q_i, 0)}{\sum_{i=1}^{L} \min(\nu_{ji} q_i, 0)} \tag{2}$$

where p and d indicate production and destruction of species, ν_{ji} represents the net stoichiometric coefficient of the jth species in the reaction i, and q_i denotes the rate of progress of the ith reaction. A negative sign has been added to the expression for normalized destruction (Eq. (2)) to emphasize that it is a consumption reaction.

The effect of rich-stage equivalence ratio on the dominant NO and NH_3 pathways for the baseline conditions is presented in Figs. 8 and 9, using normalized production and destruction contribution values. These values are calculated at a critical time instant ($\tau_g = 0.60$ ms) when the rate of change in NO concentration is highest as shown in Fig. 5(*a*). To examine the primary reaction pathways for NO_x formation and destruction, a comparison was made between $\phi_r = 1.10$ and $\phi_r = 1.80$ at $\tau_g = 0.60$ ms in Fig. 8. At the start of PFR-1 ($\tau_g = 0.60$ ms), the net NO production rate is dominated by HNO (R142, R143, R148) and Zel'dovich (R88) pathways for $\phi_r = 1.80$, whereas for $\phi_r = 1.10$ the NO_2 reaction (R180) is the main contributor. At the same instant, NO destruction is primarily driven by the Zel'dovich (R90, R88), HNO (R142), and N_2O (R84) pathways for $\phi_r = 1.10$. In contrast, for $\phi_r = 1.80$, the NH_2 and NH radicals are the main contributors to the destruction of NO concentration. To highlight the interplay between NO formation and unburned NH_3 (slippage), Fig. 9 gives the normalized NH_3 production and destruction contributions for important reactions at $\phi_r = 1.10$ and $\phi_r = 1.80$ near the start of PFR-1 ($\tau_g = 0.60$ ms). The net NH_3 production rate at $\phi_r = 1.10$ is primarily governed by the recombination reaction (R23) $NH_2 + H \rightleftharpoons NH_3$, whereas NH_2 recombination and several NH_2/N_2H_i interactions are important at $\phi_r = 1.80$. These latter NH_i/N_2H_i chemical interactions are also particularly important for determining the overall rate of reaction for ammonia combustion systems under fuel-rich conditions [19,21], as well as contribute to the relatively slow NH_3 relaxation to equilibrium shown previously in Fig. 1(*b*) for $\phi_r > 1.50$. This

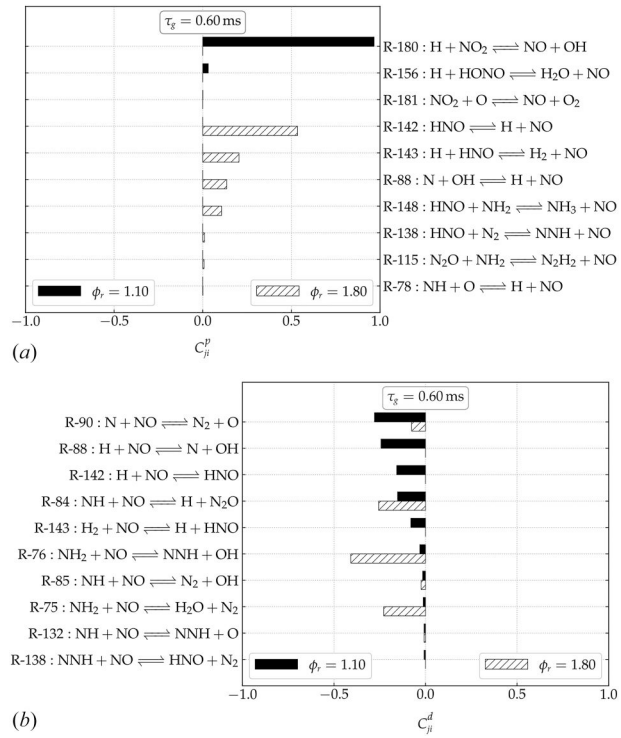

Fig. 8 Normalized NO production and destruction contributions of reactions at $\tau_g = 0.60$ ms (beginning of fuel-rich PFR-1 stage) for cases A3 and A2: (*a*) normalized NO production and (*b*) normalized NO destruction

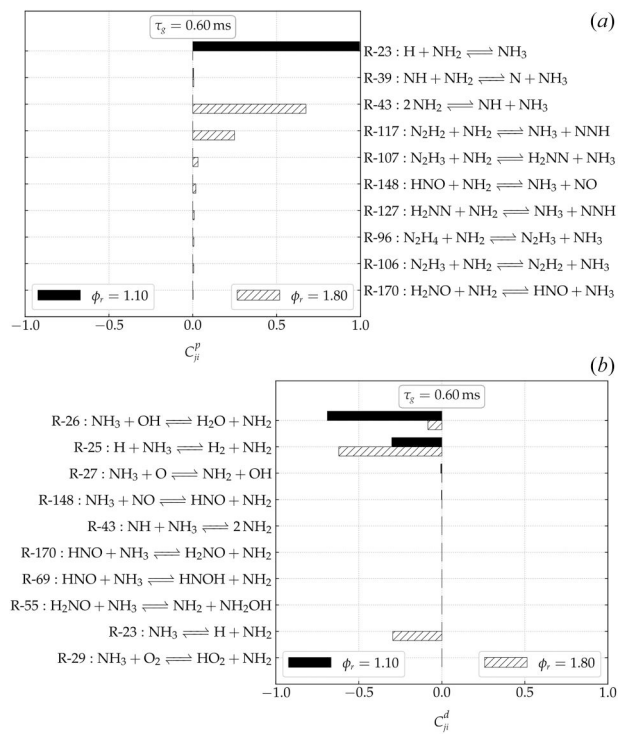

Fig. 9 Normalized NH_3 production and destruction contributions of reactions at $\tau_g = 0.60$ ms (beginning of fuel-rich PFR-1 stage) for cases A3 and A2: (*a*) normalized NH_3 production and (*b*) normalized NH_3 destruction

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increased NH_3 slippage explains the increased NO_x formation in the fuel-lean stage when air is introduced for $\phi_r > \phi_{r,\,opt}$ due to fuel NO_x pathways.

Figure 10 gives the normalized NO production and destruction contributions for important reactions in the fuel-lean stage. Comparisons were made for $\phi_r = 1.45$ and $\phi_r = 1.80$ at the exit of PSR-2 ($\tau_g = 36.50$ ms). The net NO production rate is primarily governed by the HNO (R147), Zel'dovich (R89), and NO_2 (R180, R181) pathways for both $\phi_r = 1.45$ and $\phi_r = 1.80$. In contrast, NO destruction is predominantly driven by the NO_2 (R140, R139) pathways in both cases. The abundance of the NH_2 and NH radicals becomes important to destruction of NO in the case of $\phi_r = 1.80$.

The relative importance of the major reaction pathways responsible for the production or destruction of NO is also determined through *sensitivity analysis*. This analysis, corresponding to the aforementioned critical time instants under different design conditions, is discussed in Sec. 3.5.

3.5 Sensitivity Analysis. Sensitivity analyses are used in chemical kinetic studies for identifying the rate-limiting reaction steps and to understand the relative importance of the reactions in the system. This analysis helps determine the effect of uncertainties in parameters and initial conditions on the solution of the set of ordinary differential equations commonly encountered in chemical kinetic studies. It is often convenient, for comparative analysis, to compute normalized sensitivity coefficients as [17,24–26]

$$E_{ji} = \frac{\alpha_i}{[\mathrm{M}_j]} \cdot \frac{\partial[\mathrm{M}_j]}{\partial \alpha_i} = \frac{\partial \ln[\mathrm{M}_j]}{\partial \ln \alpha_i} \tag{3}$$

The first-order elementary sensitivities in Eq. (3) represent the percentage change in a species concentration ($[\mathrm{M}_j]$) resulting from a percentage change in the parameter α_i at a given τ_g. Normalized sensitivity coefficients are dimensionless and their values are independent of the units of the $[\mathrm{M}_j]$ and α_i. The interpretation of system sensitivity through the first-order elementary sensitivity coefficients is referred to as local sensitivity analysis. In this study, using CANTERA, normalized sensitivity coefficients for species NO are determined by varying the forward rate constant of the ith reaction, which inherently affects the backward rate as well due to the equilibrium relationship. This sensitivity accounts for the net effect of both forward and backward rates, as the overall rate of progress in a reversible reaction depends on both.

For the baseline conditions corresponding to simulation output shown in Fig. 5(*a*), the local sensitivity coefficients are given in Fig. 11 at two critical time instants, where the net destruction and formation rates of NO concentration are highest, respectively. Figure 11(*a*) shows that reactions involving NH_i radicals (R76, R75, R84) have large negative sensitivity coefficients for $\phi_r = 1.80$, whereas for $\phi_r = 1.10$, the sensitivity magnitude for R84 is much higher than R76 and R75. This suggests that the destruction of NO concentration is very sensitive to rate coefficients of these reactions at $\tau_g = 0.60$ ms (beginning of fuel-rich PFR-1 stage). Additionally, the Zel'dovich pathways R88 and R90 exhibit positive and negative sensitivity coefficients for both equivalence ratios, respectively, emphasizing the role of their kinetic rate constant in the production and destruction of NO. For the fuel-lean PFR-2 stage at $\tau_g = 36.50$ ms and $\phi_r = 1.80$ (Fig. 11(*b*)), NO destruction is predominantly influenced by the R84 ($NH + NO \rightleftharpoons H + N_2O$) pathway, along with reactions involving NH_i radicals (R76, R75, R85) and Zel'dovich (R90) pathway. Meanwhile, NO production is primarily governed by the Zel'dovich (R89) and R78 ($NH + O \rightleftharpoons H + NO$) pathways under the temperature/species conditions at the end of PSR-2. At the same time instant for fuel-lean PFR-2 stage and $\phi_r = 1.45$, NO production is sensitive to the reaction rate constant of the N_2O (R190), Zel'dovich (R90), and NH (R84) pathways. In contrast, NO destruction is primarily influenced by the three-body NO_2 pathway (R140).

The sensitivity coefficients for NO formation at the end of the fuel-lean PFR-2 stage, corresponding to simulations conducted under different exit temperature conditions shown in Fig. 6, are presented in Fig. 12. For both temperature values (1800 K and

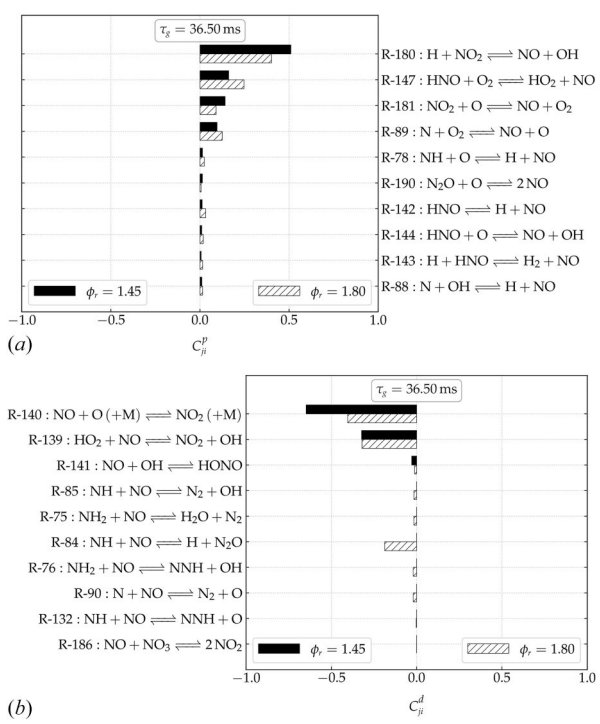

Fig. 10 Normalized NO production and destruction contributions of reactions at $\tau_g = 36.50$ ms (end of fuel-lean PSR-2 stage) for cases A1 and A2: (*a*) normalized NO production and (*b*) normalized NO destruction

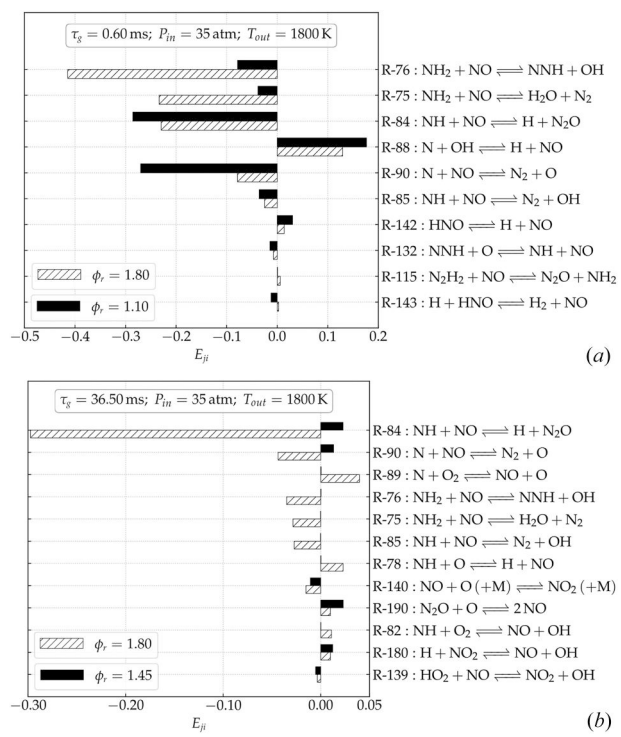

Fig. 11 Normalized sensitivity coefficients for NO for (*a*) fuel-rich stage of cases A3 and A2 and (*b*) fuel-lean stage of cases A1 and A2. (*a*) Fuel-rich stage at $\tau_g = 0.60$ ms (beginning of PFR-1) and (*b*) fuel-lean stage at $\tau_g = 36.50$ ms (end of PSR-2).

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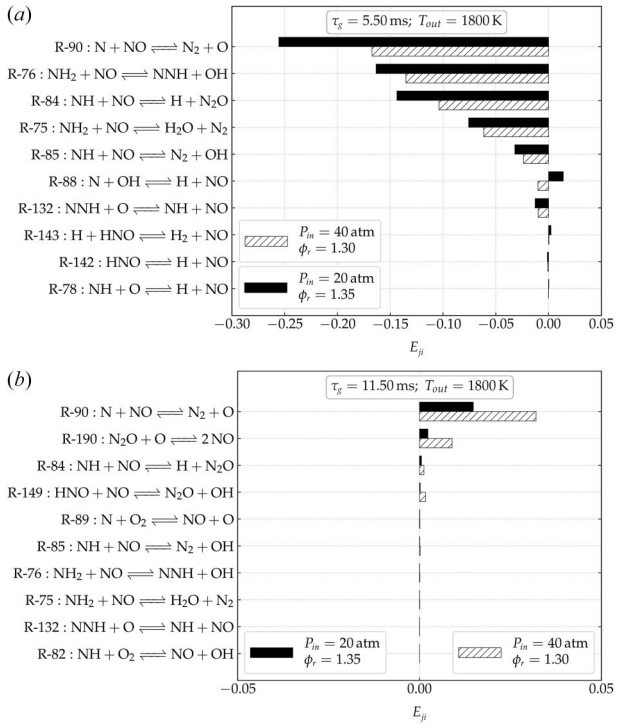

Fig. 12 Normalized sensitivity coefficients for NO for different exit temperatures at τ_g=41.50 ms (end of fuel-lean PFR-2 stage) for cases C2 and D1

1900 K), the Zel'dovich pathway (R90) exhibits the highest positive sensitivity coefficients, with the values at 1900 K being significantly larger than those at 1800 K. Thus, the rate constant for the high activation energy step ($N_2 + O \rightleftharpoons NO + N$) in the Zel'dovich mechanism plays a critical role in NO production under the given thermodynamic conditions in the fuel-lean PFR-2 stage. The same trend of higher sensitivity values at elevated temperatures is observed for N_2O and HNO reactions (R190, R149, R84, R85) as well.

The sensitivity coefficients for NO production and destruction at the end of the fuel-rich PFR-1 and fuel-lean PFR-2 stages, corresponding to simulations conducted under different design pressure conditions shown in Fig. 7, are presented in Figs. 13 and 14. The results correspond to the respective $\phi_{r,\,opt}$ for cases E1 and G1 with shorter $\tau_{PFR-1} = 5$ ms (Fig. 13), as well as for cases E2 and G2 with long $\tau_{PFR-1} = 35$ ms (Fig. 14). At the end of the fuel-rich PFR-1 stage for short τ_{PFR-1}, Fig. 13(*a*) shows that NH_i radical pathways (R76, R84, R75, R85) and the Zel'dovich pathway (R90) exhibit negative sensitivity coefficients for NO. These sensitivity

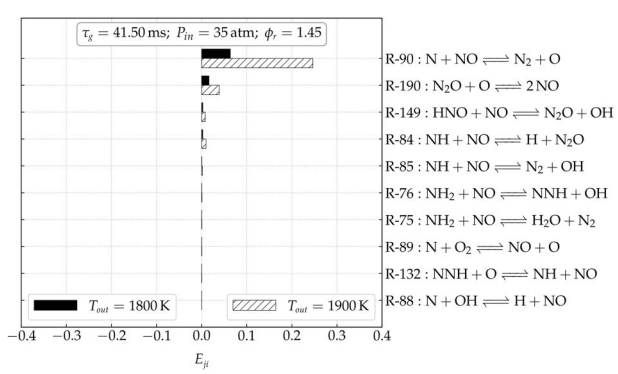

Fig. 13 Normalized sensitivity coefficients for NO at different design pressures for short τ_{PFR-1} of 5 ms for cases E1 and G1: (*a*) fuel-rich stage at τ_g=5.50 ms (end of PFR-1) and (*b*) fuel-lean stage at τ_g=11.50 ms (end of PFR-2)

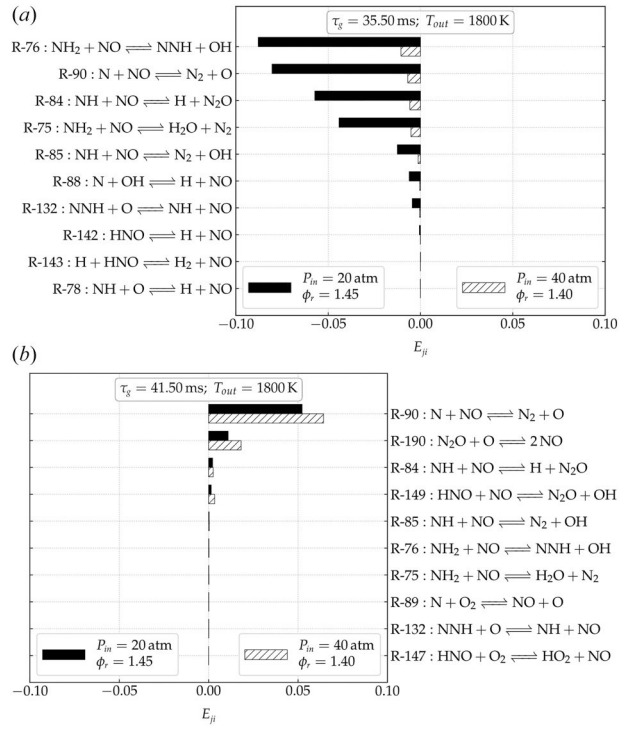

Fig. 14 Normalized sensitivity coefficients for NO at different design pressures for long τ_{PFR-1} of 35 ms for cases E2 and G2: (*a*) fuel-rich stage at τ_g=35.50 ms (end of PFR-1) and (*b*) fuel-lean stage at τ_g=41.50 ms (end of PFR-2)

coefficients are comparable at $P_{in} = 20$ atm and $P_{in} = 40$ atm. In contrast, for long τ_{PFR-1}, Fig. 14(*a*) illustrates that NH_i radical pathways (R76, R84, R75, R85) along with the Zel'dovich pathway (R90) exhibit larger negative sensitivity coefficients at $P_{in} = 20$ atm compared to $P_{in} = 40$ atm, highlighting the greater influence of their rate constants on NO formation under the lower pressure condition.

At the end of fuel-lean PFR-2 stage, Figs. 13(*b*) and 14(*b*) show that, for both short and long τ_{PFR-1}, the Zel'dovich pathway (R90) exhibits the highest positive sensitivity coefficients for both 20 atm and 40 atm pressure conditions. For short τ_{PFR-1}, the relative proportions for NO_x at combustor outlet for the design pressure of 20 atm are given by $NO : N_2O : NO_2 = 94.62\,\% : 4.64\,\% : 0.74\,\%$, whereas the corresponding percentages for 40 atm are $93.3\,\% : 5.7\,\% : 1.0\,\%$. Similarly, for long τ_{PFR-1}, the relative proportions for NO_x at combustor exit for the design pressure of 20 atm are given by $NO : N_2O : NO_2 = 97.8\,\% : 0.8\,\% : 1.4\,\%$, whereas the corresponding percentages for 40 atm are $96.0\,\% : 1.1\,\% : 2.9\,\%$. This suggests that pressure variation may not significantly impact relative proportions of the NO_x constituents for a given τ_{PFR-1}.

4 Conclusion

The NO_x emission characteristics of an NH_3–air RQL-staged gas turbine combustor were evaluated using CRN modeling across a range of conditions representative of modern gas turbines, including variations in pressure, exit temperature, and fuel-rich stage equivalence ratio. The simulations identified the optimal stoichiometry and residence times for each stage to minimize NO_x emissions while also quantifying the effects of pressure and temperature. Reaction pathway and sensitivity analyses were conducted to identify the key elementary reaction steps involved in NO_x formation and reduction for the given reaction mechanism. Our results show that ammonia combustion with minimal emissions is theoretically achievable and can enable zero-carbon power generation at low NO_x emissions levels (e.g., < 20 ppm) meeting site requirements with zero or minimal selective catalytic reduction cleanup.

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The results of this study highlight that the residence time in PFR-1, exit temperature, rich-stage equivalence ratio, design pressure, and residence time in PFR-2 are the primary factors influencing NO_x emissions at the combustor outlet. Simulations under varying exit temperature conditions indicate that selecting the residence time in PFR-2 is crucial for reducing overall NO_x emissions for a given τ_{PFR-1}. For a given exit temperature, NO_x concentrations exhibit a slight initial decline, followed by an increase, as the residence time of PFR-2 increases. The rise in NO_x concentrations with increasing PFR-2 residence time is more pronounced at higher exit temperatures. The effectiveness of high pressure in varying NO_x emissions also depends on the τ_{PFR-1}. For short τ_{PFR-1}, the emission levels are significantly higher, and the optimal-NO_x decreases with increasing design pressure. On the other hand, for long τ_{PFR-1}, the emissions are substantially lower, and NO_x increases slightly with pressure, but the difference in optimal-NO_x concentrations between the studied design pressures is minimal. For baseline conditions, NO_x production/consumption are mainly controlled by HNO, NO_2, N_2O, and Zel'dovich pathways, with a notable contribution from NH_i radical reactions, particularly at higher values of ϕ_r. Sensitivity analysis highlights the importance of the kinetic rate constants of reactions involving NH_i radicals and Zel'dovich pathways under gas turbine-relevant conditions for predicting overall NO_x emissions at the exit.

Acknowledgment

This material is based upon work supported by the Department of Energy under Award No. DE-FE0032169.

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Funding Data

- U.S. Department of Energy (DOE Award No. DE-FE0032169; Funder ID: 10.13039/100000015).

Data Availability Statement

The datasets generated and supporting the findings of this article are obtainable from the corresponding author upon reasonable request.

Nomenclature

C_{ji}^{d} = normalized destruction of the jth species of the ith reaction
C_{ji}^{p} = normalized production of the jth species of the ith reaction
E_{ji} = normalized sensitivity coefficient of the jth species of the ith reaction
P_{in} = inlet pressure (atm)
T_{in} = inlet temperature (K)
T_{out} = outlet temperature (K)

Greek Symbols

τ = residence time (ms)
ϕ_l = lean-stage equivalence ratio
ϕ_r = rich-stage equivalence ratio

Superscripts and Subscripts

d = destruction
g = global
in = inlet
ji = jth species of the ith reaction
l = lean
opt = optimum
out = outlet
p = production
r = rich

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